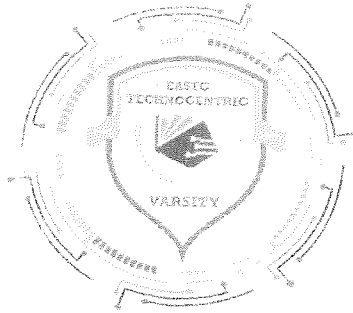


LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 03	Electricity and Electronics, NQF Level 4.	13



38
40
95%

ELECTRICITY AND ELECTRONICS
FORMATIVE ASSESSMENT

Module #	671101000-KM-03:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	Munyisi Ndabeni
Contact Address	433 Tswelopele Section, Tembisa
Telephone (H)	0838004724
Telephone (W)	
Facsimile	
Cellular	0838004724
E-mail	mvndabeni@gmail.com

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <i>Munyisi Ndabeni</i>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Munyisi Ndabeni*

Learners Name: *Munyisi Ndabeni*Learner's signature: *[Signature]*

Date: _____

Facilitator Name: _____

Facilitator's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____

Date: _____

Employee witness: _____

Date: _____



PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Munzir Ndabeni</i>	Signature <i>[Signature]</i>
Facilitator Name:	Signature:

[Faint signature]

Complete the following questions as per instructions provided

LEARNING UNIT 1:

KM-03-KT01: Fundamentals of electricity (35%)

Learning Outcome 1:	Principles and fundamental concepts of electricity
Learning Outcome 2:	Definitions, types, properties and applications of conductors, insulators and semi-conductors
Learning Outcome 3:	Concepts, theories and principles of Electrical Circuits
Learning Outcome 4:	Calculations on basic electrical circuits (resistance, voltage and current) and power
Learning Outcome 5:	Basic principles and calculation of magnetism

Task	Questions Description	Marks
1.	a. Describe, calculate and interpret fundamental concepts of electricity (Electro motive force, Potential difference, Resistance) using the correct units of measurement and definitions. b. Describe the factors that influences the resistance of a material c. Define and explain, using the correct units of measurement ohms law of electricity d. Define and explain using the correct units of measurement Kirchoff's law of electricity Manipulate formula to calculate voltage, current and resistance in series/parallel circuits	10

Electromotive force, potential difference and resistance are fundamental electrical concepts defined by energy, charge, and opposition to flow. Electromotive force (EMF) is the total energy per unit charge supplied by a source (Volts.V), P.D is energy dissipated between two points (Volts.V), and resistance is the opposition

to charge flow (Ohms). They are linked by Ohm's law ($V = IR$) determining circuit behavior.

(b) Length: Resistance is directly proportional to the length of the conductor. A longer wire provides more area for collisions increasing resistance.

Temperature: For most metals, resistance increases with high temperatures due to increased atomic vibrations causing more collisions with electrons.

Cross-sectional Area: Resistance is inversely proportional to the area. A thicker wire (large cross-sectional area) provides more space for electrons to flow decreasing resistance.

Material: The inherent properties of the material, such as its atomic structure, determine how easily it conducts electricity.

Impurities: The presence of impurities in the material can also impact the overall resistance.

(c) Ohm's law defines the relationship between Voltage (V), Current (I), and resistance (R) in an electrical circuit.

Stating that current is directly proportional to Voltage and inversely proportional to resistance. It uses Volts, Amperes (A), and Ohms (Ω) to calculate circuit values.

(d) Kirchhoff's law are fundamental circuit analysis rules based on conservation of charge and energy. Current law states total current entering a node equals that leaving, measured in Amperes (A). Voltage law states the sum of potential differences around any closed loop is zero, measured in Volts.

Series Circuit: $I_{total} = I_1 = I_2 = I_n$ Current, Voltage $V_{total} = V_1 + V_2 + V_n$
Resistance $R_{total} = R_1 + R_2 + \dots$

Parallel circuit: Current $I_{total} = I_1 + I_2 + \dots$, Voltage $V_{total} = V_1 = V_2 = V_n$

Resistance $\frac{1}{R_{total}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

Task	Questions Description	Marks
2	a) Describe, with the aid of drawings, the effect on the magnetic field around a current carrying conductor when placed in a uniform magnetic field. Fleming's left-hand rule must be demonstrated. b) Describe and explain the various forces or methods that can be used to alter magnetic fields, and describe the changes that take place	10

a) When a current-carrying conductor is placed in a uniform magnetic field, its circular magnetic field interacts with the external field, causing compression of field lines on one side and stretching on the other, resulting in a force on the conductor. The fields add together on one side (stronger field) and cancel on the other (weaker field), creating a resultant field that forces the conductor to move. Fleming's left-hand rule: predicts the direction of the force. If the forefinger points in the direction of the magnetic field (B) and the middle finger in direction of the current (I), the thumb points in the direction of the force (F). The force F on a conductor of length L with current I in field B is given by $F = ILB \sin \theta$

b) Magnetic fields are altered by changing electric currents, moving magnets or altering magnetic materials. Key methods include increasing current (strengthening the field), using iron cores (concentrating flux), changing distances, or reversing current polarity.

Changing electric current (electromagnets): Increasing the current in a coil increases the magnetic field strength.

Reversing the current direction reverses the magnetic field polarity.

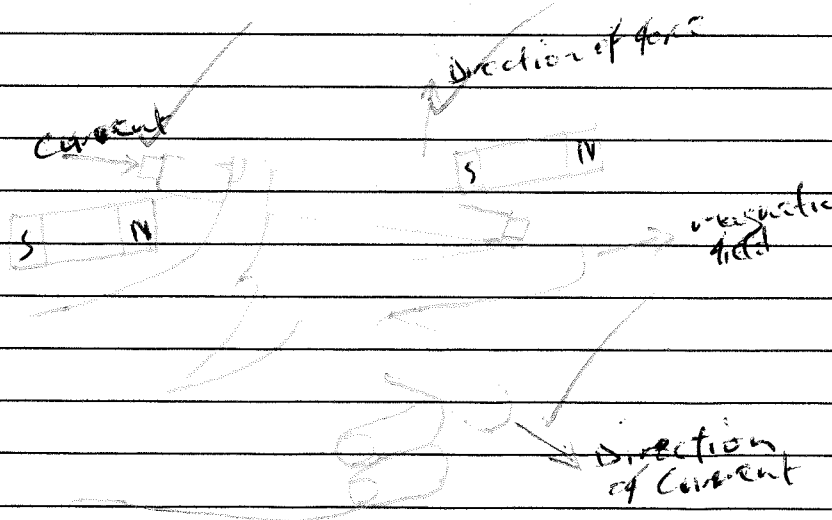
Introducing ferromagnetic materials: Placing materials like iron inside a coil concentrates the magnetic flux, significantly strengthening the magnetic field.

Motion of magnets/coils: Moving a permanent or electro magnet closer increases the magnetic field strength at a point while moving away decreases it.

Changing the number of turns: Increasing the number of turns in a solenoid (coil) increases the magnetic field strength.

Temperature changes: Heating a magnet can reduce its magnetic strength by disrupting the alignment of its magnetic domains.

Magnetic shielding: Placing a high permeability material (like mu-metal) around a sensitive area can redirect magnetic field lines, shielding the interior from the external field.



LEARNING UNIT 2:

KM-03-KT03: Earthing and bonding (35%)

Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
3	a) Explain the regulatory requirements for the earthing of neutral conductors on both the supplier and consumer side of an installation b) Explain the term 'systems earthing' c) Describe the earthing systems relevant to SANS 10142-1. d) Describe how a common earth electrode is used in reticulation circuits	10

a) Supply

Source grounding: The neutral conductor of the supply transformer must be directly connected to the earth to establish a consistent reference voltage.

Protective multiple earthing: The supplier often connects the neutral to earth at multiple points along the distribution line, often providing a combined protective earth and neutral conductor.

Maintenance: Suppliers must maintain earthing connections, ensuring resistance meets safety standards.
Safety limits: Earthing connections must be designed to withstand potential fault currents without over heating.

Consumers

● **Main earthing terminal:** The consumer must connect the neutral conductor to the main earthing terminal at the service panel.

Separation: Downstream from the main switch/service entrance, the neutral and earth conductors must remain separated to prevent neutral current from flowing through earth paths, except in specific PEN systems.

Bonding: All metal work in the installation must be bonded to the earthing system.

Alternative Supplies: If a generator is used, the neutral must be earthed and switched appropriately to prevent dangerous back feeding of the grid.

(b) System earthing is the intentional connection of an electrical network's neutral point (or phase conductor) to the ground, providing a safe, low-impedance path for fault currents to return to the source. System earthing ensures that in the event of a fault, excessive current flows through a controlled path, immediately activating protective devices like MCB.

(c) SANS 10142-1 governs low-voltage electrical installations in South Africa, primarily mandating the TN-C-S (Protective Multiple Earthing-PME) system, where the supply neutral is earthed at the source and at intervals, splitting into separate neutral and earth conductors at the consumer's point of supply. The standard ensure safe, low-impedance paths for fault currents, requires proper bonding of all metallic parts, and governs neutral-earth bonding for alternative supplies.

d) Safety function: The electrode connects the neutral point of a transformer or metallic, non-current-carrying parts of equipment to the ground, prevent dangerous voltage buildup on exposed metal surfaces.

Installation types: commonly used in articulation system as vertical rods/pipes driven into the ground, horizontal tapes, plates, or a foundation ring earthing system surrounding a structure for maximum, consistent contact with soil moisture:

Low resistance requirement: To be effective, the electrode must reach subsoil moisture and often requires interconnection to maintain a low-resistance path.

Location: They are frequently located at substations, distribution poles, or as part of the earthing system at the consumer's main distribution board to provide a common reference.

LEARNING UNIT 2:

KM-03-KT02: Electronics (30%)

Learning Outcome 1:	International Electro technical Commission (IEC) symbols for electronic components
Learning Outcome 2:	Principles, safety precautions, identification and basic function of electronic components
Learning Outcome 3:	Basic electronic circuits
Learning Outcome 4:	Principles of rectification, single phase and 3-phase

Task	Questions Description	Marks
4	a) Describe the various capacitors used and typical applications b) Determine the value of capacitors using charts and calculate the value of capacitance of capacitors in series and parallel	10
	c) Determine the value of resistors utilizing colour coding charts	

(a) Ceramic Capacitors: Ideal for high-frequency applications, RF circuits, decoupling (noise bypass) in ICs, and general-purpose electronics.

Electrolytic capacitors: Power supply filtering (smoothing), bulk energy storage, and low-frequency applications.

Film capacitors: Radio signal paths, motor run/start, power factor correction, and precision circuits.

Tantalum capacitors: Timing, filtering and bypass in space-constrained, high-reliability application.

Supercapacitors (ultracapacitors): Short-term energy storage, memory backup, and regenerative braking systems.

(d) Capacitors in Series

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \quad C_1 = 6\mu F \quad C_2 = 3\mu F$$
$$\frac{1}{C_T} = \frac{1}{6} + \frac{1}{3}$$

$$\frac{1}{C_T} = \frac{1}{6} + \frac{2}{6} = \frac{3}{6}$$

$$C_T = 2\mu F$$

Rule: Total capacitance decrease in series

Capacitors in parallel

$$C_T = C_1 + C_2 + C_3$$
$$= 6\mu F + 3\mu F = 9\mu F$$

Total capacitance increases in parallel

e) Brown - Black - Red - Gold

Brown = 1

Black = 0

Red = $\times 100$

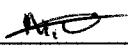
Gold = $\pm 5\%$

SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1				
2				
3				
4				


Learner Signature

Date


Facilitators' Signature

Date

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

Good work

ASSESSMENT JUDGEMENT

Learner's Total Mark:
not met ☐

Requirements met ☒

Requirements

Action/s required:

By when:

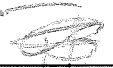
LEARNER FEEDBACK & COMMENTS

Good work

DECLARATION BY THE FACILITATOR

I Baennoah Tyabachi (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, <u>Mangisi Ndabani</u>declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
			
Learner	Date	Facilitator	Date

DECLARATION BY THE ASSESSOR			
I <u>Buennah Tjabadi</u> (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
			
Assessor	Date	Moderator	Date

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes

What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method.

1. Written Assignment : Printed Questionnaire
2. Observation : As per Practical Assessment Guide
3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation

What resources would you require to conduct the assessment?

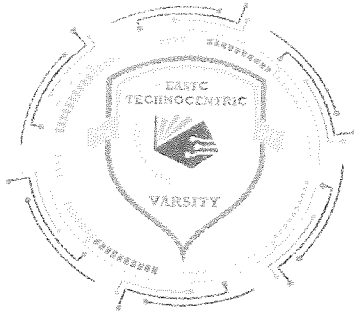
1. Assessor toolkit
2. Theoretical assignment
3. Practical assessment
4. Learner toolkit

How would you sequence your activities? Explain

1. Record preparation process
2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment
3. Learner will write summative assessments

4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit





55
60
92%

ELECTRICITY AND ELECTRONICS
SUMMATIVE ASSESMENTS

Module #	671101000-KM-03:
NQF Level	level 4,
Notional hours	130
Credit(s)	Cr 13
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

m

Details	Please Complete this Section
Name	Ilwajisi Mdaabani
Contact Address	
Telephone (H)	
Telephone (W)	
Facsimile	
Cellular	0835004724
E-mail	mdaabani@gmail.com

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).



LEARNER ASSESSMENT PLAN

Unit code	TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
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1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <i>Munyisi Ndabeni</i>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Munyisi Ndabeni*

Learners Name: *Munyisi Ndabeni*Learner's signature: *[Signature]* Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
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3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
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ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Munyisi Ndabeni</i>	Signature: <i>[Signature]</i>
Assessors Name: <i>Baleenah Jabadi</i>	Signature: <i>[Signature]</i>

[Faint signature]

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INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.

LEARNING UNIT 1:

KM-03-KT01: Fundamentals of electricity (35%)

Learning Outcome 1:	Principles and fundamental concepts of electricity
Learning Outcome 2:	Definitions, types, properties and applications of conductors, insulators and semi-conductors
Learning Outcome 3:	Concepts, theories and principles of Electrical Circuits
Learning Outcome 4:	Calculations on basic electrical circuits (resistance, voltage and current) and power
Learning Outcome 5:	Basic principles and calculation of magnetism

Task	Questions Description	Marks
1	List types of materials used for conductors, insulators and semi-conductors and describe their mechanical and electrical properties and applications.	10

Conductors

Materials: Copper (most versatile, high conductivity),
Aluminium (lightweight, used in overhead power lines),

silver (highest conductivity, expensive), gold (corrosion-resistant), iron, steel and graphite.

Electrical properties: low electrical resistivity,
high electrical conductivity, and low resistance.

Mechanical properties: Generally high tensile strength,
ductility, and malleability, allowing them to be drawn
into thin wires.

Applications: Wiring in home devices, power transmission
lines, motor winding and circuit boards.

Insulators

Materials: Rubber, glass, plastic, ceramic, dry wood,
mica, paper and teflon.

Electrical properties: High resistivity, preventing electrical leakage.

Mechanical properties: Frequently rigid, brittle, and non-ductile.

Applications: Cable insulation, circuit board substrate (fiberglass), insulating coatings, and insulators in high voltage transmission towers.

Semi-conductors

Semi-conductors have electrical conductivity between conductors and insulators, controllable via doping.

Materials: Silicon (most common, inexpensive), germanium and gallium arsenide (used in high speed devices).

Electrical properties: Conductivity increases with temperature or impurities.

Task	Questions Description	Marks
2	a) Name, describe and explain the different types of magnets and their properties b) Explain fundamental magnetic concepts by naming the five characteristics of magnetic lines of force and explaining the relationship between flux and flux density	10

Permanent magnets

Description: Materials that create their own, persistent magnetic field without external power, such as magnets on a fridge or bar and horseshoe magnets.

Properties: High coercivity (hard to demagnetize) and high remanence (strong magnetism remaining after magnetization).

Temporary magnets

Description: Materials that act as magnets only when within a strong magnetic field and lose their magnetism when the field is removed.

Properties: Low coercivity, high permeability (easily magnetized).

Electromagnets

Description: Magnets created by passing an electric current through coil of wire (solenoid), often with a ferromagnetic core.

Properties: Controllable strength, can be turned on and off, can be extremely strong, requiring continuous energy.

Characteristics

Continuous closed loops: They form unbroken loops, travelling from the north pole to the south pole externally and returning from south to north inside the magnet.

Directional: They possess a specific direction, conventionally flowing from the north pole to the south pole.

Non-intersection: Magnetic lines of force never cross or intersect one another.

Density indicates strength: The closer the lines are together, the stronger the magnetic field; sparse lines indicate a weaker field.

Lateral repulsion and longitudinal contraction: Lines repel each other side-by-side and try to contract longitudinally.

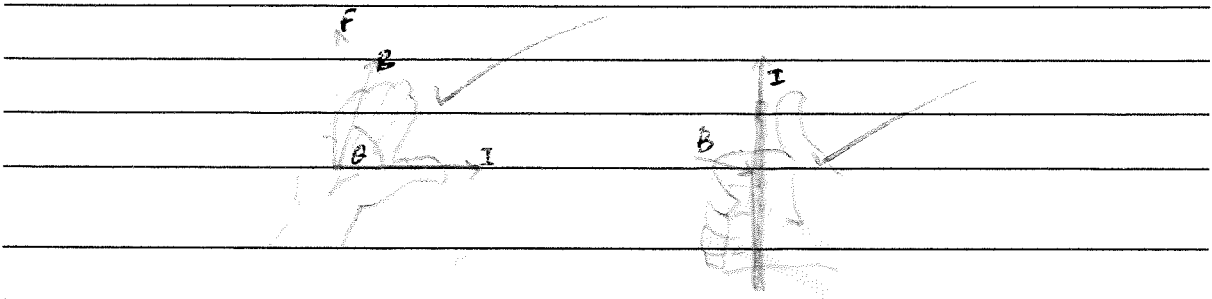
Relationship

Magnetic flux (Φ): This is the total, overall quantity of magnetic field lines produced by a source.

Magnetic flux density (B): This is the concentration of that magnetic flux passing through a specific unit area.

The relationship: The density of the flux is directly proportional to the total flux and inversely proportional to the area (A) through which it passes.

Task	Questions Description	Marks
3	Describe, with the aid of a drawing, the application of the right-hand grip rule and show how a magnetic field is established when an electrical current flows through a conductor, by using formula, calculate the force on a current carrying conductor.	10



The right-hand grip rule determines the direction of the magnetic field (B) around a conductor carrying conventional current (I). By pointing the right thumb in the current direction, the curled fingers indicate the circular field direction, as illustrated below. The force on the conductor is calculated by:

$$F = BIL \sin \theta$$

Application of the right-hand grip rule

Grip the conductor with your right hand.

Point the thumb in the direction of the electric current.

Curl your fingers around the conductor. The direction in which your fingers point represents the direction of the magnetic field lines.

$$F = BIL \sin \theta$$

$$= 2 \text{ T} \times 10 \text{ A} \times 0.5 \text{ m} \times 1$$

$$F = 10 \text{ N}$$

~~$\sin \theta$~~

$$\sin \theta = \sin 90 = 1$$

$$L = 0.5 \text{ m}$$

$$B = 2 \text{ T}$$

$$I = 10 \text{ A}$$

$$F = ?$$

LEARNING UNIT 2:

KM-03-KT03: Earthing and bonding (35%)

Learning Outcome 1:	KT0301 Principles of earthing and bonding
Learning Outcome 2:	KT0302 Methods of earthing and bonding on low voltage overhead lines, equipment and cables.
Learning Outcome 3:	KT0303 Regulatory and statutory requirements related to earthing and bonding

Task	Questions Description	Marks
4	a) Define earthing and bonding and explain the purpose thereof b) Explain the regulatory requirements for earthing and bonding	10

a) Earthing

Definition: The process of connecting the metallic, non-current-carrying parts of electrical equipment or the neutral point of a supply system to the earth via a low-impedance conductor.

Purpose: Provide a safe path for fault current to ground, causing protective devices (breaker/fuse) to trip immediately, preventing electric shocks.

Bonding

Definition: The practice of connecting all exposed, non-current-carrying metallic parts (e.g. pipes, structural steel, conduits) of an electrical installation together to create a continuous, low-resistance path.

Purpose: Prevents arcing or sparking between metal components by eliminating potential difference.

Provides a path back to the main earthing terminal to trigger protective devices if a fault occurs.

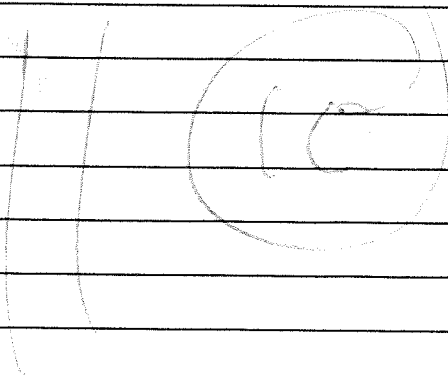
(b) Earthing system: A grounding electrode system (e.g. rods, plates) must connect to the consumer's earth terminal to facilitate the automatic disconnection of supply.

Main bonding: Main equipotential bonding must connect all extraneous conductive parts (incoming water/gas services) to the main earthing terminal (METS).

Supplementary bonding: Additional bonding is required in areas with electrical shock hazards to ensure all exposed and extraneous conductive parts are at the same potential.

Conductor sizing: Conductors must be sized to handle fault currents (e.g. copper with a cross sectional area of at least 25mm^2 in certain PME systems).

Inspection: Regular testing (e.g. continuity test & R_s & R_e earth resistance tests) and documentation of results are required to ensure ongoing safety compliance.



Task	Questions Description	Marks
5	a) Describe the provision of earthing for underground cables and overhead lines b) Explain the various processes of measuring, testing and calculating earthing and bonding system values	10

(a) Earthing for underground cables

Metallic sheath/Armor earthing: The metallic sheath or armor of a cable is connected to earth at both ends (double-ended earthing) or at a single point (single point earthing) to prevent circulating currents while maintaining a continuous path for fault current.

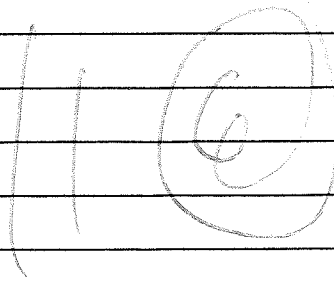
Cable joints and terminations: Proper grounding is required at cable joint boxes to ensure the sheath remains at earth potential.

Earthing for overhead lines

Earth wire (shield wire): An earth wire made of galvanized steel is installed at the highest point of tower/poles, providing a direct low-resistance path to protect against lightning.

(b) These processes include measuring earth resistance

using a ground resistance tester, testing continuity of bonding conductors and assessing the effectiveness of the earthing system by comparing the measured values.



LEARNING UNIT 3:

KM-03-KT02: Electronics (30%)

Learning Outcome 1:	International Electro technical Commission (IEC) symbols for electronic components
Learning Outcome 2:	Principles, safety precautions, identification and basic function of electronic components
Learning Outcome 3:	Basic electronic circuits
Learning Outcome 4:	Principles of rectification, single phase and 3-phase

Task	Questions Description	Marks
6	a) Describe the various inductors used and typical applications b) Describe the construction and uses of various semiconductor devices c) Draw basic electronic circuits using International Electrotechnical Commission (IEC) symbols	10

air core inductors: Utilize magnetic ceramic materials with

air core inductors: Use non-magnetic cores (air/plastic), offering high Q factor but lower inductance. Used in high-frequency applications like radio antennas, RF circuits, and oscillator networks.

ferrite core inductors: Utilize magnetic ceramic materials with high permeability. Ideal for high-frequency, high-performance applications like SMPS (Switch mode power Supplies) and RF circuits.

Iron core inductors: Utilize a solid iron core, providing high inductance but higher core losses. Used in low-frequency, high-power scenarios like audio equipment and power supply filters.

Toroidal inductors: Have a continuous closed-shaped (toroid) core. Their closed magnetic loop reduces electromagnetic interference (EMI), making them perfect for compact power supplies and audio equipment.

Variable inductors: Feature an adjustable core to change the inductance value, used in tuning circuits for radio, oscillator circuits and more.

Multilayer chip inductors: Surface mount devices (SMD) with stacked layers, ideal for high-density, compact electronic circuits.

Coupled inductors: Two or more coils sharing a magnetic core, used to isolate circuits, boost/buck voltage, and act as transformers in power supply.

b) Diodes (P-n Junction)

Construction: Created by joining p-type and n-type semiconductors, creating a junction that allows current flow in only one direction.

Uses: Rectifying AC to DC voltage regulation, and protecting circuits as described on the RS components.

Transistors (BJT/FET)

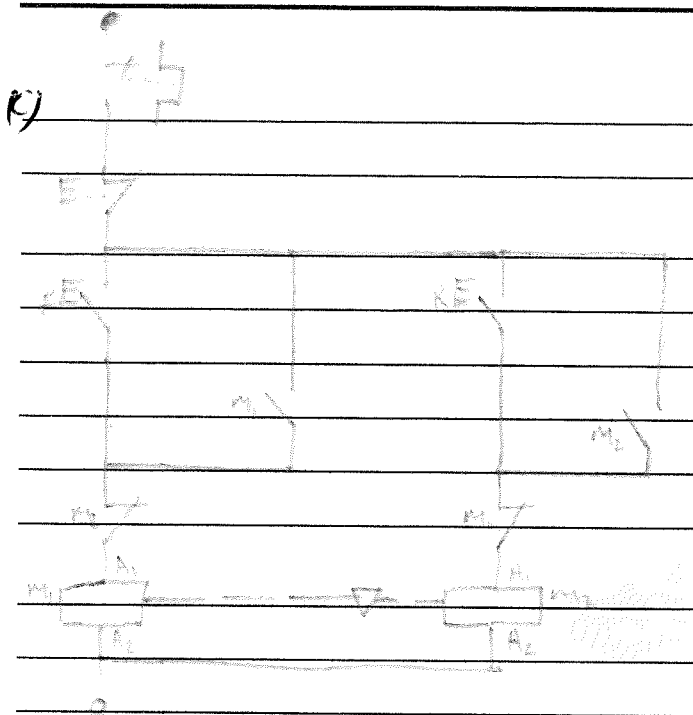
Construction: Formed by sandwiching p-type between two n-types (NPN) or n-type between two p-types (PNP) to create terminals - emitter, base, and collector. FETs use an electric field to control conductivity.

Uses: Acting as fast electronic switches for data processing or as amplifiers for signal processing.

Integrated circuits (IC)

Construction: Thousands to billions of transistors, diodes,

and capacitors are built onto a single microscopic silicon wafer using photolithography, extending to uses! The "brains" of electronics, used in microprocessors, memory chips (RAM, flash) and computer systems.



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Assessor's Report

Candidate's Name	Munyis Ndabeni	ID No.	950115587089
Assessor's Name		Reg. No.	
Title	Electricity and Electronics		
ASSESSMENT DECISION			
Outcome	C	NYC	Comments
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			
			
Signature of Assessor		Signature of Candidate	

A handwritten signature in black ink, appearing to be "M".

Candidate Feedback Report

Candidate's Name	Mungisi Ndabeni		ID No.	950115587087
Assessor's Name		Reg. No.		
Title	Electricity and Electronics			
ASSESSMENT DECISION				
Source of Evidence	C	NYC	Comment	
Knowledge questions	✓		Good	
Product /Work sample	✓		Good	
Project	✓		Good	
Indirect Evidence	✓		Good	
Overall Assessment Decision				
Additional Notes				
Date				
				
Signature of Assessor			Signature of Candidate	

Candidate Appeal Form

Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Title:		
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

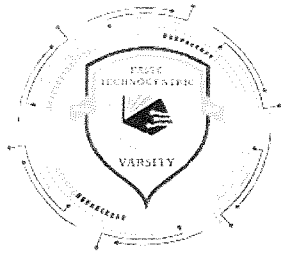
Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Muyisi Ndabeni	ID No.	9501115587089
Title	Electricity and Electronics		
MODERATION DECISION			
Outcome	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

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ASSESSMENT REVIEW

NAME of LEARNER <i>Munyisi</i>		NAME of ASSESSOR	
VENUE <i>EASTC</i>		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
Learner	Date	Assessor	Date
		Moderator	Date



LEARNER FILE DIVIDER

CONTENT	CHECKED
LEARNER WORKBOOK	
SUMMATIVE ASSESSMENT	

KNOWLEDGE MODULES

KM ID	TITLE	Credits
KM – 04	Industry Context, NQF Level 3.	2

Mrugisi Ndabeni



24 24 = 80
30
80

INDUSTRY CONTEXT

SUMMATIVE ASSESMENT



Module #	671101000-KM-04:
NQF Level	level 3,
Notional hours	20
Credit(s)	Cr 2
Occupation Specialisation or	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

Details	Please Complete this Section
Name	Munyisi Ndebele
Contact Address	433 Tswelopele section, Tembisa
Telephone (H)	
Telephone (W)	
Facsimile	
Cellular	0838004724
E-mail	munyisi.ndebele@gmail.com

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Welcome to your Summative Assessment!

This document deals with the summative assessment of your competence towards the outcomes which you need to achieve. "Summative Assessment is assessment for making a judgement about achievement. This is carried out when a learner is ready to be assessed at the end of a programme of learning" (SAQA: Criteria and Guidelines for Assessment Policy Document,).

LEARNER ASSESSMENT PLAN

Unit code	Module title	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language
Resources
Barriers

People to be involved with assessment

Learner: <i>Munyisi Ndabeni</i>	Manager:
Trainer:	Mentor / Coach:
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Munyisi Ndabeni*

Learners Name: *Munyisi Ndabeni*Learner's signature: *N/A* Date: _____

Assessors Name: _____

Assessor's signature: _____ Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

20



Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____

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PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓

Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Mungisi Ndabeni</i>	Signature <i>[Signature]</i>
Assessors Name:	Signature:

[Faint signature]

INSTRUCTIONS:

The following sections will test your knowledge and you are required to answer all the questions.

All true and false items must be answered.

Once you have finished you may obtain feedback from your assessor on the accuracy of your answer.



LEARNING UNIT 1:

KM-04-KT01: Introduction to the world of work and the electrical trade (100%)

Learning Outcome 1:	KT0101 The Electricians world of work
Learning Outcome 2:	KT0102 Career opportunities of qualified Electricians
Learning Outcome 3:	KT0103 The Electrician's responsibilities and duties
Learning Outcome 4:	KT0104 Legislation relating to apprentices in the Electrical trade
Learning Outcome 5:	KT0105 Trade test requirements

Task	Questions Description	Marks
1	Describe and explain the job environment and workplace roles of an Electrician are accurately	10

Electricians install, maintain, and repair electrical power, lighting and control systems in residential, commercial, and industrial settings. The work environment is physically demanding, involving tasks in confined spaces (Attics, crawl spaces), at height, or on noisy construction sites, requiring strict adherence to safety protocols like IPEC codes.

Job Environment

5 Varied Settings: Work occurs in homes, office buildings, schools, plants, factories and construction sites.

Workplace roles

Installation: Reading blueprints and technical diagrams to install wiring, circuit breakers and lighting systems in new or renovated buildings.

Task	Questions Description	Marks
2	Describe the profile of an Electrician with respect to industry descriptions, career path progression and requirements	10

Industry descriptions

Residential: Work in homes, installing wiring, outlets and lighting.

Commercial: Work in office buildings, schools, and hospitals, dealing with larger electrical systems, fire alarms, and security.

Industrial: Maintain heavy machinery, motors, and high-voltage systems in factories and power plants.

Inside wireman: Focus on installing electrical conduit and system within buildings.

Outside lineman: Work on high-voltage power line and transmission systems.

Career path progression

Apprentice: The starting point, involving paid on-the-job training and technical classroom instruction to learn regulations and safety.

Journeyman Electrician: After completing the apprenticeship and passing a licensing exam, they work independently, diagnose issues and perform installations.

Master electrician: The top level, requiring roughly 7 years of experience and about 6 complex licensing exams.

Requirements and skills: Education (high school diploma or equivalent is required), Licensing (Apprentice, journeyman, or master)

Key skills (Strong troubleshooting abilities, knowledge of IVEC).

9

Task	Questions Description	Marks
3	Explain Trade test methodology, requirements and assessment procedure leading up to the trade assessment	10

A trade test is a final integrated summative for artisan qualification, conducted by QCTO accredited centres to evaluate technical proficiency.

Key Trade test requirements

Documentation and eligibility: Applicants must provide a C.V, I.D, service letters detailing at least 3 years of work experience, and equal educational qualifications.

ARPL assessment: Before the main test, ARPL is used to assess prior learning, identifying skill gaps in candidates who may not have formal training but possess relevant work experiences.

Trade test preparation course: This is a crucial step that ensures candidates are ready to face the summative assessment through mock assessments, reviewing theoretical knowledge and improving technical skills.

Booking: Once requirements are met, an application is submitted to an accredited trade test centre.

Assessment procedure

Application and approval: Accredited trade test centres must issue a test date with 5 working days of application.

Test Day protocols: The test consists of practical tasks. Candidates are expected to manage time, follow safety regulations, and properly select materials and tools (e.g. in electrical tests, selecting the correct meter for a sequence).

Moderation and Certification The assessment is conducted by a NAMB-registered assessor. Results are submitted to NAMB, and upon passing the CPTC issues a certificate within 21 working days.

②

Assessor's Report

Candidate's Name	Mungisa Ndabeni	ID No.	9501115587089
Assessor's Name	Baqerrah Tyabadi	Reg. No.	NAMB-A/ 2025/5307
ASSESSMENT DECISION			
	C	NYC	Comments
OVERALL COMPETENCE JUDGEMENT:			
Comments			
Date			
			
Signature of Assessor		Signature of Candidate	



Candidate Feedback Report

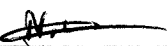
Candidate's Name	Munji, Ndabeni	ID No.	950115587089
Assessor's Name		Reg. No.	
Unit Standard Title	Industry context		
ASSESSMENT DECISION			
Source of Evidence	C	NYC	Comment
Knowledge questions	✓		Good
Product /Work sample	✓		Good
Project	✓		Good
Indirect Evidence	✓		Good
Overall Assessment Decision			
Additional Notes			
Date			
			
Signature of Assessor		Signature of Candidate	

M

Candidate Appeal Form

Candidate's Name:	ID No.	
Assessor's Name:	Reg. No.	
Date:		
SECTION 1		
Candidate's reason for disagreeing with the assessment decision		
Assessor's rationale for the assessment decision		
Candidate's signature		
Assessor's signature		
SECTION 2		
Moderator's decision		
Moderator's signature		

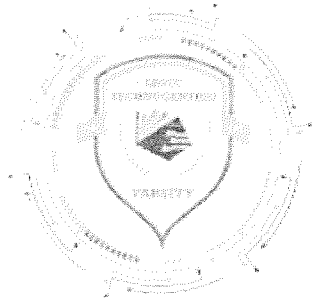
Moderator's Report

Moderator's Name		Reg. No.	
Assessor's Name		Reg. No.	
Candidate's Name	Munyisi Ndabeni	ID No.	9501115587089
MODERATION DECISION			
	C	NYC	Comments
Overall Moderation Decision			
Feedback to Assessor			
Action Required			
Date of Moderation			
Signature of Moderator			
Signature of Assessor			
Signature of Candidate			

ASSESSMENT REVIEW

NAME of LEARNER <i>Mirayisa</i>		NAME of ASSESSOR	
VENUE <i>EASTC</i>		DATE of REVIEW	
Review Dimension	ASSESSOR	LEARNER/ CANDIDATE	ACTION
The principles/criteria for good assessment were achieved?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment related to the registered unit standard?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment was practical?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
It was time efficient and cost-effective and did not interfere with my normal responsibilities?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment instruments were fair, clear and understandable	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The assessment judgements was made against set requirements	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The venue and equipment was functional?	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Special needs were identified and the assessment plan was adjusted	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
Feedback was constructive against the evidence required	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
An opportunity to appeal was given	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
The evidence was recorded	☐ Agree ☐ Disagree	☒ Agree ☐ Disagree	
LEARNER'S DECLARATION OF UNDERSTANDING			
I am aware of the moderation process and understand that the moderator could declare the assessment decision invalid			
Learner	Date	Assessor	Date
Moderator		Date	

M. Ndabeni



22/30
25
887

INDUSTRY CONTEXT
FORMATIVE ASSESSMENT

4/15

Module #	671101000-KM-04:
NQF Level	level 3,
Notional hours	20
Credit(s)	Cr 2
Occupation or Specialisation	671101000: Occupational Certificate: Electrician
Associated Occupation	671101: Electrician

m

Details	Please Complete this Section
Name	Mvuyisi Mlabeni
Contact Address	
Telephone (H)	
Telephone (W)	
Facsimile	
Cellular	0838004724
E-mail	mvndabeni@gmail.com

About the Formative Assessment

Purpose	The purpose of this Formative Assessment is to evaluate learners understanding on the specific outcomes and/or assessment criteria and prepare the learner for the summative assessment. This will show that the learner is ready to be assessed.
Context	This assessment represents the Formative Assessment component and should be completed in the classroom/training room.
Resources	The following are resources needed for this assessment: 1. Learner Guide; and 2. Assessment Preparation.
Instructions to Facilitators	Facilitators will be required to: • Explain the completion of the Formative Assessment to each learner; and • Interview the learner on similar questions, should he/she not be able to write.
Instructions to Learners	Learners will be required to: • Complete the Formative Assessment as per the instructions; • Ensure that all questions are completed; • Ensure that the completion of the workbook is their own work; • Ensure that all annexure are attached to the workbook and clearly referred to;
Assessment Time	Learners are required to complete this assessment within the allocated time frame/hours.
Total Mark	This formative assessment carries a total mark of _ points . In order to meet the pass mark, learners are required to achieve a minimum of 80% of the total marks.

LEARNER ASSESSMENT PLAN

Unit code	/MODULE TITLES	NQF level	Credits	✓

Activity	Evidence of activity will be found where	Place & planned date of activity	Date Completed
Training	Classroom training registers	Training Provider Date:	
Self assessment	Assessment contract signed & dated	Learner file Date:	
Assessment contract	Assessment contract signed & dated	Learner file Date:	
Initial meeting	Assessor briefing checklist	Learner file Date:	
	Assessment contract & assessment plan	Learner file Date:	
Formative Assessment	Assessment instruments	Learner file Date:	
Summative Assessment	Assessment instruments	Learner file Date:	
Other Evidence	Research portfolio (if applicable)	Learner file Date:	
Feedback	Feedback Report	Learner file Date:	
Moderation	Moderators report	Learner file Date:	
Judgement	Assessor Summary Report / Moderator report	Learner file Date:	
1 st Reassessment	Assessors summary report / instruments	Learner file Date:	
2 nd Reassessment	Assessors summary report / instruments	Learner file Date:	

Special arrangements for assessment

Place
Language

Resources
Barriers

People to be involved with assessment

Learner: <i>Munyisi Ndabeni</i>	Manager:
Trainer:	Mentor / Coach:
Facilitator	
Assessor:	Moderator:

Next steps for learning

Resources required for this assessment

Guidelines to the learner: *Munyisi Ndabeni*

Learners Name: *Munyisi Ndabeni*

Learner's signature: *[Signature]*

Date: _____

Facilitator Name: _____

Facilitator's signature: _____

Date: _____

ASSESSMENT APPEALS PROCEDURE

1. A learner has the right to appeal under the following circumstances
 - If the laid down assessment procedures were not followed during assessments
 - If not all evidence available was taken into account during the assessment
 - The assessor was not a subject matter expert or did not have a subject matter expert during the assessment process
 - The assessor did not assess according to the performance criteria and range statement stipulated in the unit standard
 - Not all the range items were available for assessment
2. A learner bringing an appeal against a decision of the assessment will lodge such an appeal with the assessor and the internal moderator within 2 days of the assessment feedback session.
3. A learner bringing an appeal should complete the "Learner's Notice of Assessment Appeal" form before the Appeal Hearing. The form should be handed to the internal moderator or a representative of the SETA.
4. Should the internal moderator re-affirm the assessor's decision, the learner may appeal to the external verifier within 2 days after the initial moderator's feedback session. The external verifier's decision will be final. Should the external verifier re-affirm the assessors' decision, the cost for re-evaluation will be upon the learner. Should the verifier's decision differ from the assessor's decision, the cost for re-evaluation will be borne by the assessor.

ASSESSMENT APPEAL APPLICATION FORM

LEARNER'S NOTICE OF ASSESSMENT APPEAL

TO: The Internal Moderator

A meeting with the internal moderator is hereby requested to discuss the outcome of my assessment.

Internal moderator name _____

Date of submission: _____

Name of employee assessed: _____

Name of Assessor: _____

Date of feedback session: _____

Grounds for Appeal

No	Tick the applicable ground(s) for appeal	Tick
1	The assessment did not follow the laid down procedure	
2	Not all evidence available was taken into account during the assessment	
3	The assessor was not a subject matter expert nor was a subject matter expert present during the assessment process	
4	The assessment was not according to the performance criteria and the range statement stipulated in the unit standard	
5	Not all the range items were available for the assessment	

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QW

Reasons for Appeal

No	Please give detailed reasons for the choice(s) above
1	
2	
3	
4	
5	

Learner's signature: _____ Date: _____

Employee witness: _____ Date: _____



PRE-ASSESSMENT MEETING CHECKLIST

Points Assessor must cover in the initial meeting with the learner - Please tick

Item	Points to be covered	Tick
1	Welcome the candidate and put them at ease	✓
2	Explain the purpose of the meeting (why you are there and how long the meeting will take)	✓
3	Explain the ▪ NQF ▪ Credits ▪ Certification process ▪ Learning pathways	✓
4	Explain ▪ Who is involved in the assessment and their role (learners, coach, assessors, managers, moderators) ▪ Principles of assessment (fairness, confidentiality, validity, sufficiency)	✓
5	Explain the assessment process? ▪ Check learner readiness for assessment (logbook / self assessment) ▪ Assessment contract to be completed ▪ Preparation of learner (this meeting) ▪ The assessment (observation and knowledge questionnaire) ▪ Judgement of the evidence ▪ Outcome of assessment (competent, not yet competent, need further evidence)	✓
6	Give Learner copies of the following documentation and explain each document ▪ The Assessment Guide which includes ○ The relevant unit standard (s) ○ Assessment contract ○ Assessment plan ○ Observation checklist ○ Knowledge checklist	✓
7	Discuss the assessment plan (complete the assessment plan document) ▪ Allow the learner to participate in the decisions made ▪ Agree on dates, time and venue for the assessment and feedback ▪ Agree on evidence the learner can submit ▪ Agree and explain the assessment methods ▪ Identify and discuss special assessment needs of the candidate ▪ Identify and eliminate unfair barriers (language, disability etc) ▪ Discuss and agree on witness requirements	✓
8	Tell the candidate his/her rights and responsibilities, the assessment procedures and policies ▪ How many times they may be assessed ▪ Appeals process / procedure ▪ Reassessment policy	✓
9	Ensure the assessment environment is appropriate or make special arrangements	✓
10	Discuss moderation	✓
11	Allow the learner opportunity to clarify any items discussed	✓



Learner declaration of acceptance of assessment instruments and relevant documentation: Date:	
Learners Name: <i>Munyisi Ndabane</i>	Signature: <i>[Signature]</i>
Facilitator Name:	Signature:

[Faint signature]

Complete the following questions as per instructions provided

LEARNING UNIT 1:

KM-04-KT01: Introduction to the world of work and the electrical trade (100%)

Learning Outcome 1:	KT0101 The Electricians world of work
Learning Outcome 2:	KT0102 Career opportunities of qualified Electricians
Learning Outcome 3:	KT0103 The Electrician's responsibilities and duties
Learning Outcome 4:	KT0104 Legislation relating to apprentices in the Electrical trade
Learning Outcome 5:	KT0105 Trade test requirements

Task	Questions Description	Marks
1.	Explain the Legal aspects that includes the employment contract and roles and responsibilities of the employer and employee	10

Legal aspects

Essential contract elements: Must include employer/employee details, job description, start date, salary, working hours, overtime policies and notice periods.

Statutory compliance: Contracts cannot provide lesser rights than the basic conditions of employment Act (BCEA), such as statutory leave entitlements.

Roles and responsibilities of the employer

Provide work and remunerate as agreed.

Ensure a safe, healthy working environment per the OHS Act.

Respect employee rights (confidentiality, privacy).

Role and Responsibilities of the employee:

Perform agreed tasks efficiently and abide by workplace policies.

Adhere to confidentiality and restraint of trade clauses.

Comply with working hours and provide notice for absenteeism.

⑧ Types of contracts: Includes permanent, fixed-term, seasonal, and temporary agreements, which determine benefits and security.

Dispute Resolution: Disputes can be referred to the CCH or internal company procedures fail.

Task	Questions Description	Marks
2	Describe the use of applicable legislation relating to the electrical trade	5

Applicable legislation in the electrical trade, such as the occupational health and safety act and electrical installation regulations, mandate safe work practices, mandatory certifications, and proper installation standards.

Aspects of electrical legislation usage

Safety standards (CHSA/CNHN): legislation sets strict requirements for safe electrical installation, maintenance, and equipment use, mandating the use of personal protective equipment (PPE) and safe procedures, such as isolating and locking out equipment.

Installation standards: Defines the minimum requirements for design, installation and testing, such as proper grounding, correct sizing and using approved components, ensuring compliance and preventing hazards.

Licensing and certification: Only registered electrical contractors are authorized to perform work, ensuring technical expertise.

Certificate of compliance (COC): Legislation dictates that a COC must be issued by a registered person upon completion of electrical work, proving it safe and compliant for property transfers.

Legislation enforcement: Non-compliance can lead to penalties, including fines or imprisonment for hazardous work.

Task	Questions Description	Marks
3	Explain Trade test methodology, requirements and assessment procedure leading up to the trade assessment	10

A trade test is a final integrated summative assessment for artisan qualification, conducted by QCTO accredited centres to evaluate technical proficiency.

Trade test requirements and preparatory steps

Documentation and eligibility: Applicants must provide a C.v, I.D, service letters detailing at least years of work experience, and educational qualifications.

ARPL assessment: Before the main test, ARPL is used to assess prior learning, identifying skill gaps in candidates who may not have formal training but possess relevant work experience.

Trade test preparation course: This is a crucial step that ensures candidates are ready to face the summative assessment through mock assessments, reviewing theoretical knowledge and improving technical skills.

Booking: Once requirements are met, an application is submitted to an accredited Trade Test Centre.

Assessment procedure

Application and approval: Accredited Trade Test Centre must issue a test date within 5 working days of application.

Test day protocols: The test consists of practical tasks.

Candidates are expected to manage time, follow safety regulations, and properly select materials and tools (e.g. in electrical tests, selecting the correct order for sequence)

Q

11 Moderation and certification. The assessment is conducted by a NAMB-registered assessor. Results are submitted to NAMB and upon passing the QCTO issues a certificate within 21 working days.

The trade test ensures that the candidate is proficient, competent, and ready for work, and it is a key step towards becoming a qualified artisan.

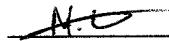
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SELF-ASSESSMENT

The learner must make use of the following self-evaluation checklist to rate himself against the learning outcomes of this particular training module in establishing the level of mastery of the information.

1. Not able to comply
2. Reasonable compliance (Not acceptable for final evaluation)
3. Able to comply fully

OUTCOMES		1	2	3
1				
2				
3				
4				



Learner Signature

Date

Facilitators' Signature

Date

ASSESSMENT FEEDBACK REPORT

FACILITATOR FEEDBACK & REMARKS

ASSESSMENT JUDGEMENT

Learner's Total Mark:
not met ☐

Requirements met ☐

Requirements

Action/s required:

By when:

[Signature]

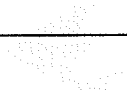
LEARNER FEEDBACK & COMMENTS

Good work

DECLARATION BY THE FACILITATOR

I (Facilitator) hereby certify that I have examined the learner workbook and I am satisfied with the evidence provided by the learner.

DECLARATION BY LEARNER			
I, <u>Mungisi Maleni</u>declare that I am satisfied that the feedback given to me by the Facilitator was relevant, sufficient and done in a constructive manner. I accept the assessment judgment and have no further questions relating to this particular assessment event.			
			
Learner	Date	Facilitator	Date

DECLARATION BY THE ASSESSOR			
I (Assessor) hereby certify that I have examined the learner workbook and I am satisfied with the Facilitator Judgment of this assessment.			
			
Assessor	Date	Moderator	Date

Application for Assessment

First Attempt ☐ Second Attempt ☐ Third Attempt ☐ RPL ☐

Observer/Witness required: Yes/No

Name: _____

Contact Number: _____

Interpreter required: Yes/No

Name: _____

Contact Number: _____

Was Employer informed of request for Assessment: Yes/No

Type of Assessment: Theoretical

Practical

Recommendation for Assessment

The applicant has completed his/her training, has prepared for the assessment and is now ready to be assessed.

The ETDP Practitioner believe that the candidate is ready for the assessment because he has completed the relevant unit standard.

Notification of Assessment

Assessment will take place as follows:

	Theory	Practical
Date		
Time	9H00	9H00
Venue	Employer premises	Employer premises
Instruments	Written assignment	Practical Assessment

Action required from candidate:

Candidate need to prepare him/herself to either write the Questionnaire or conduct Practical Assessment in order to achieve competency towards the relevant unit standard.

ETDP PRACTITIONER

Assessment Planning

	<u>Theory</u>	<u>Practical</u>
Date of assessment:		
When would be the best time for the Assessment?	9H00	9H00

Have you received an application to assess the candidate? Yes
What assessment methods do you consider using? (observation, assignment, presentation, etc.) Explain method. 1. Written Assignment : Printed Questionnaire 2. Observation : As per Practical Assessment Guide 3. Interview : List of Questions as per Practical Assessment Guide after Practical Observation
What resources would you require to conduct the assessment? 1. Assessor toolkit 2. Theoretical assignment 3. Practical assessment 4. Learner toolkit
How would you sequence your activities? Explain 1. Record preparation process 2. Meet with group/candidate and explain assessment procedure – how you are going to conduct the assessment 3. Learner will write summative assessments



4. Conduct observation in workplace where applicable
5. Complete learner toolkit
6. Complete assessor toolkit